# IEEE Testbed UI Manual

## Contents

# 1. Abbreviations

| Abbreviation | Definition |
|---|---|
| API | Application programming interface |
| IMEI | International Mobile Equipment Identity |
| IMSI | International Mobile Subscriber Identity |
| MCC | Mobile Country Code |
| MNC | Mobile Network Code |
| NGAP | Next Generation Application Protocol |
| PCAP | Packet Capture |
| PDU | Protocol Data Unit |
| SUPI | Subscription Permanent Identifier |
| TAC | Tracking Area Code |
| UE | User Equipment |
| UI | User Interface |

# 2. Introduction

IEEE 5G/6G Innovation Testbed is an end-to-end cloud based 5G network. Testbed resources can be accessed using CLI and UI modes. This document describes the accessing of Testbed using UI mode.

# 3. About Testbed UI

IEEE Testbed UI is a JavaScript based Graphical User Interface for IEEE 5G/6G Innovation Testbed. It exposes the Testbed capabilities through APIs for simplifying testbed operations.

**Design:**

- Frontend: Angular JS
- Backend: Node JS/Express JS
- Database: MongoDB

## 3.1. Testbed UI Sections

IEEE Testbed UI contains various sections (given below) for accessing and managing each entity of the Testbed.

- Dashboard
- Users
- Testbed
- User cases & Tools
- Resources

### 3.2.1 Dashboard

Dashboard defines the overall network overview of testbed including the gNodeB and 5G core network function. It also shows the below major KPIs that defines the overall status of the testbed

- Network Overview for gNodeB and 5G core.
- KPI(s) for Users, UE(s), Monitoring and Tracing.
- Use cases and tools information

### 3.2.2 Users

This section contains the user management details.

- **Admin**: Can add, modify and delete users
- **Non admin users**: Can view the configured details and download the required key files (for ssh login).

### 3.2.3 Testbed

This section contains management of gNodeB, UEs, 5G core, Tracing and Monitoring of the Testbed.

#### 3.1.3.1 gNodeB

gNodeB is the 5G wireless base station that transmits and receives messages between the UE and the 5G core network. gNodeB establishes NGAP connectivity with the 5G Core.

- Admin can update the configuration for gNodeB (Mainly “NGAP IP address”, “GTP IP address” and “AMF IP address and port”)
- Admin can start/stop gNodeB.
- Non-admin users can view gNodeB details and can start gNodeB.

#### 3.1.3.2 UE

Users can create emulated UEs (User Equipment) and perform operations in this section.

- UEs can be added by pressing “Add UE” button.
- Once UEs are created, users can “Modify”, “Delete” or “Register” UEs in the “Configuration” tab.
- Registered UEs will appear in “Operations” tab.
- In the “Operations” tab, users can perform below options on UE.
    - View UE’s data (like IMSI, IMSI, status, timers, tunnel information etc.)
    - Establish PDU session
    - PDU session release
    - Deregister
    - Release all PDU sessions
    - Dump all UE information (to a file)

#### 3.1.3.3 5G core

The 5G Core in the Testbed is orchestrated using Kubernetes. Each network function of the 5G core is represented as a Kubernetes Pod.

In this section, users can view various aspects of the 5G core (given below) and configure subscriber information.

- System metrics (CPU Usage and Memory Usage)
- PODs List (network functions)
- Services List
- Replica List
- Deployment List
- Config Maps List
- In “Logging” tab, users can view and download the logs.

#### 3.1.3.4 Monitoring

This section provides the capability to the users for monitoring 5G core metrics using Grafana and 5G core network functions using tools like Linkerd and Kubernetes dashboard.

#### 3.1.3.5 Tracing

This section provides the capability to the users for capturing PCAP logs for the test scenarios.

### 3.2.4 Use cases and Tools – Under development

This section provides various tools and provisions for different uses cases. Below is the list of tools and use cases supported in this section.

- CAMARA API Integration.
- Network Slicing
- Linux Traffic Control Integration.
- Open Speed Test
- Trace Analyzer
- DDoS Attack

### 3.2.5 Resources

This section provides resources for different uses cases and features available in the Testbed. Resources include the list of documents and videos for different uses cases and features

# 4. Accessing Testbed UI

This section explains the steps required for accessing the IEEE Testbed UI.

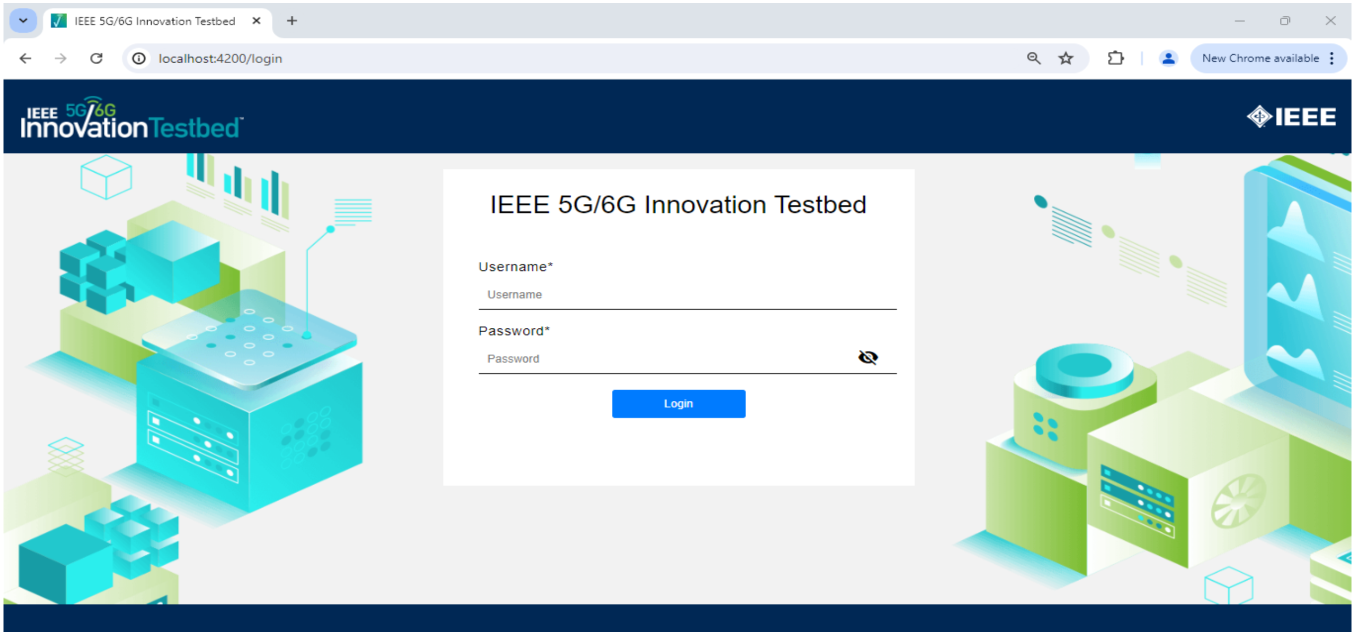
**Step-1**: **Connect to UERANSIM EC2 instance using forwarding method.**
IEEE Testbed UI runs on the standard port 4200 for frontend and 8400 for backend support.
For accessing UI, users need to connect to UERANSIM EC2 instance using the port forwarding method.

- *ssh -I <key_file> username@<ip-address> -p 3022 -L 4200:127.0.0.1:4200 -L 8400:127.0.0.1:8400*

Example: *ssh -i "ieee-aws-ssh-key.pem" ubuntu@13.233.101.116 -p 3022 -L 4200:127.0.0.1:4200 -L 8400:127.0.0.1:8400*

**Step-2**: **Login to UI in any browser using the below URL.**

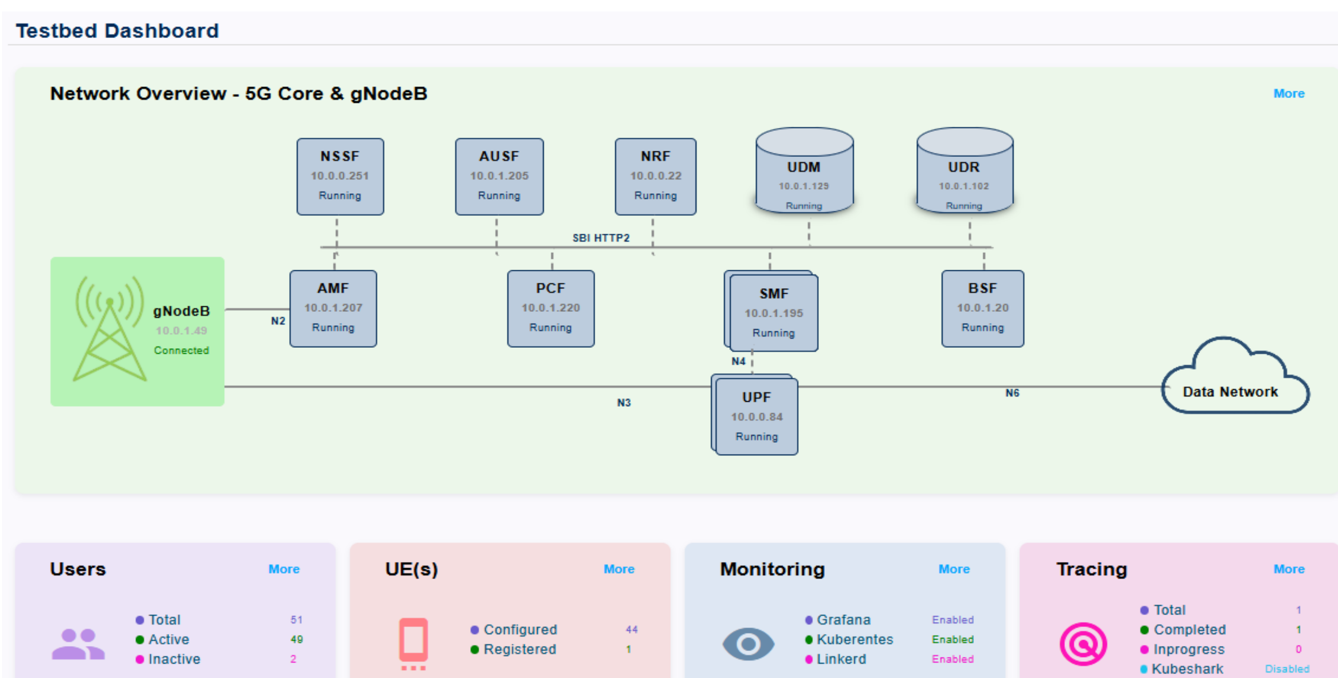
- ***http://localhost:4200/login***

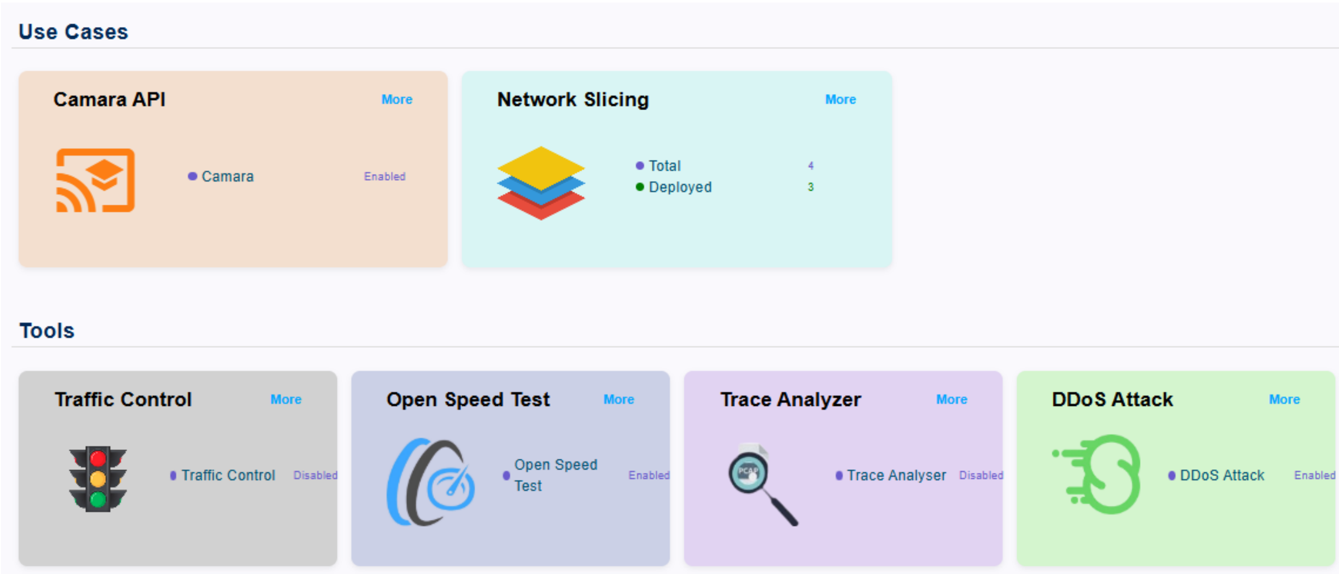



Once a user enters the credentials (username and password), he or she will be redirected to the dashboard page

## 4.1 Dashboard

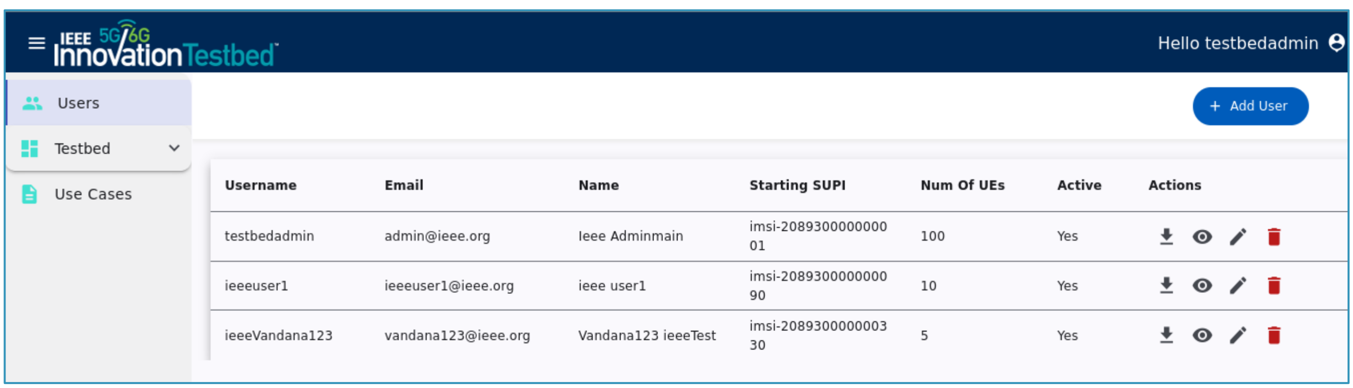
This section defines the dashboard containing overall network overview of testbed including the gNodeB and 5G core network function, KPI(s) for Users, UE(s), Monitoring, Tracing & Use cases and tools information






## 4.2 User management

Administrator can add more users in the "User" section. Also, admin can delete users and modify user data.



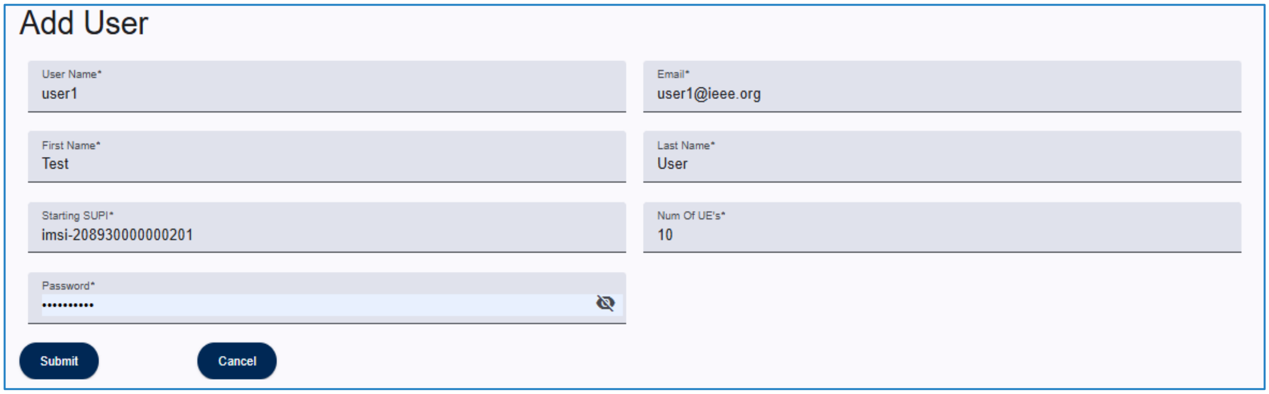

### 4.2.1 Adding a new user

For adding a user, admin can click "Add User" button and fill the below.

- Username: Unique username
- Email
- First name
- Last Name
- Starting SUPI: IEEE provides the IMSI ranges for managing the IMSI numbers for the users. This is just a recommended range. Admin can specify custom ranges for different users. Below are the important guidelines for the admin while adding IMSI numbers for users.
  1. IMSI numbers should be of 15-digit length.

2. IMSI numbers should be prefixed with “imsi-“.
3. IMSI numbers/ranges should not overlap with IMSI numbers/ranges of other users.

- Number of UEs: Value provided in this filed is used to create UEs for the users starting from IMSI number specified in the filed “Starting SUPI”.
- Password




### 4.2.2 Editing User details

Admin can edit user’s password by clicking edit button under “Actions” column.



## 4.3 Testbed management

In this section, users can get information about accessing gNodeB, UE, 5G core, Tracing and Monitoring of the Testbed.

### 4.2.1 gNodeB

This section contains the sub-sections below.

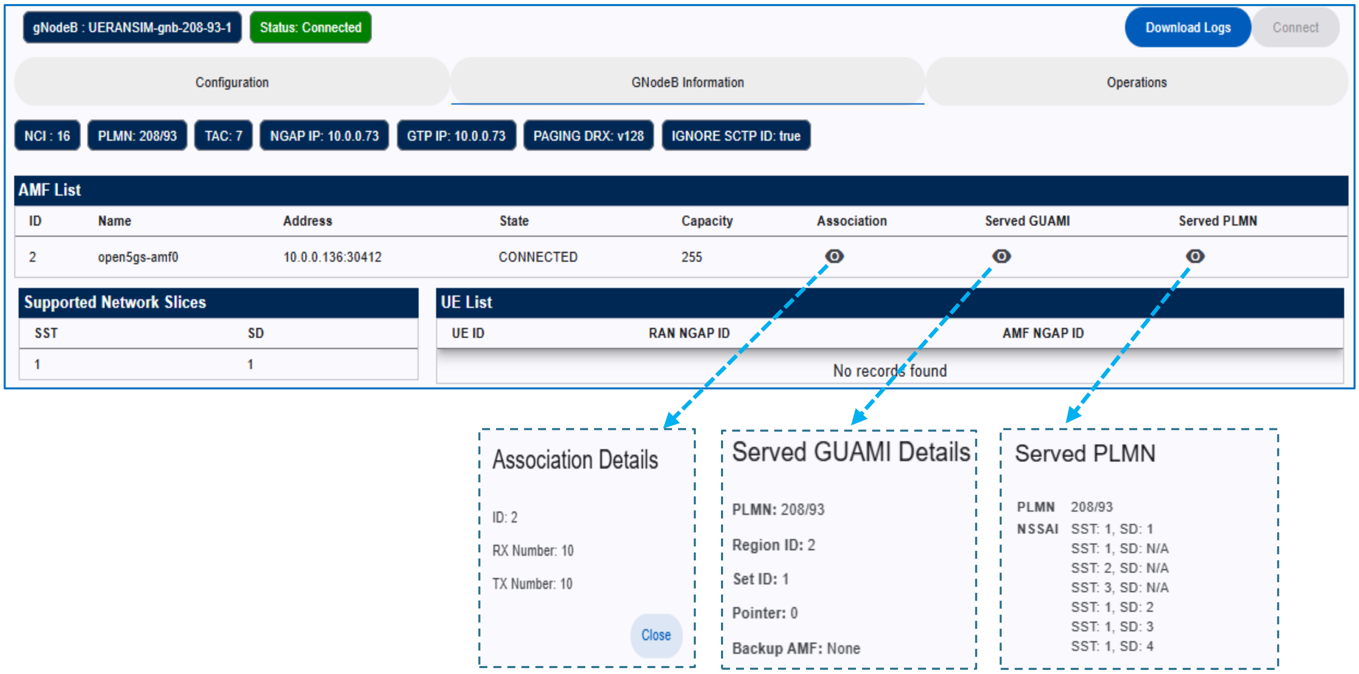
- Configuration
- Information
- Operations

#### 4.3.1.1 Configuration

Admin can update “MCC”, “MNC”, “TAC”, “NGAP IP Address", “GTP IP Address”, “AMF IP Address”, “Port” and Slice configuration.


#### 4.3.1.2 Information

This section will be updated when gNodeB is in connected state.



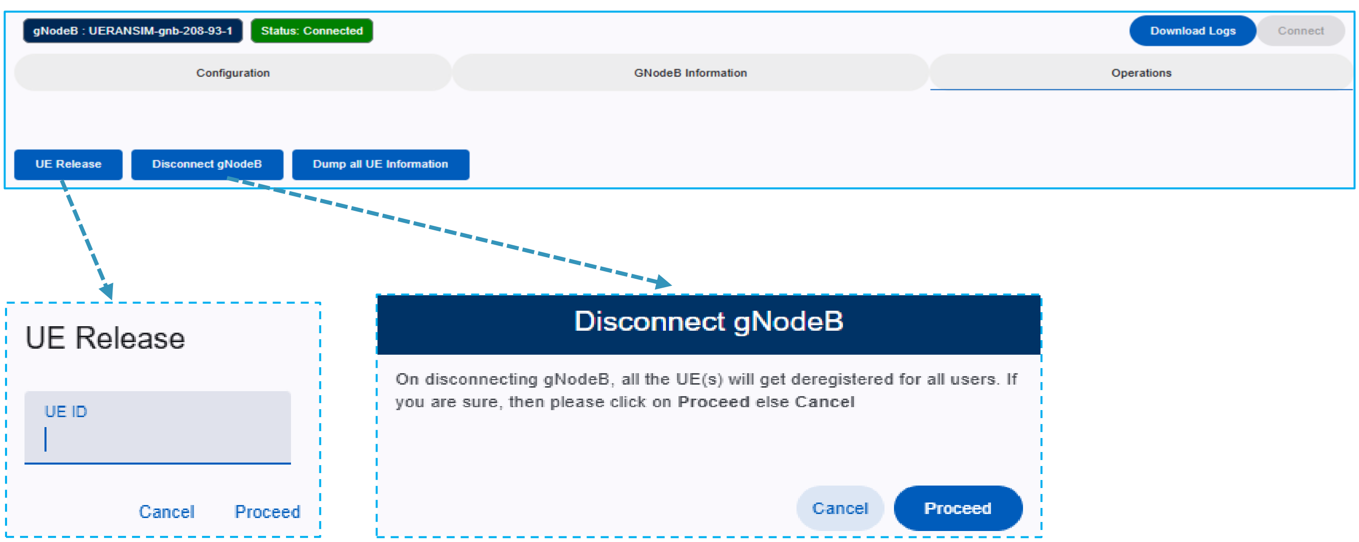

### 4.3.1.3 Operations

Various gNodeB operations are supported in this section.

UE Release: Release NGAP connection of the UE

Disconnect: gNodeB will be disconnected from 5G core and all UEs will be deregistered (if any).

Download gNodeB Information: gNodeB Information will be downloaded to a local file.



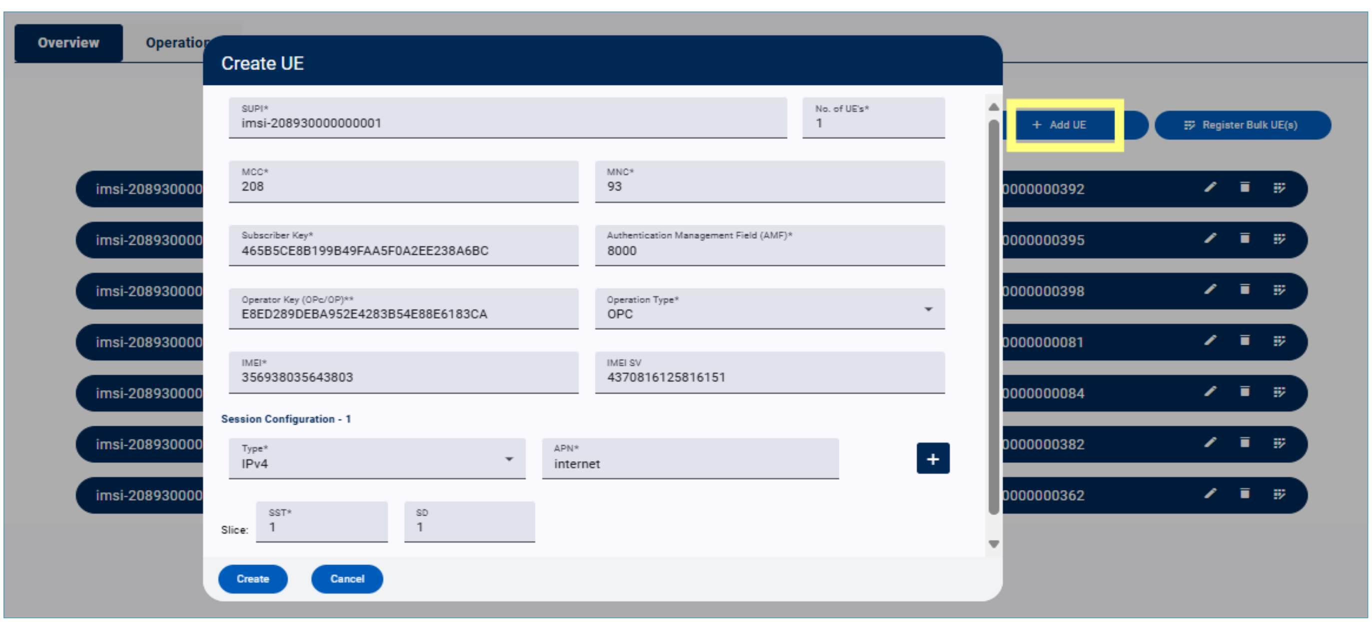
## 4.3.2 UE

This section contains the list of UEs configured for a particular user.

### 4.3.2.1 Configuration

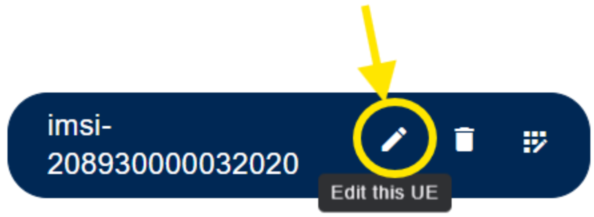

#### 4.3.2.1.1 Add UE

User can add emulated UEs using “Add UE” button.
The “SUPI” field should be added with the allocated IMSI number (by Admin in User section) and “No. of UE’s” can be updated with required number of UEs. Rest of the fields can leave with default values.



#### 4.3.2.1.2 Edit UE

User can update the UE details using the "Edit" button.




#### 4.3.2.1.3 Delete UE

User can delete UE entry using the "Delete" button.


#### 4.3.2.1.4 Register UE

Users can register the UE to 5G network by using the "Register UE" button.

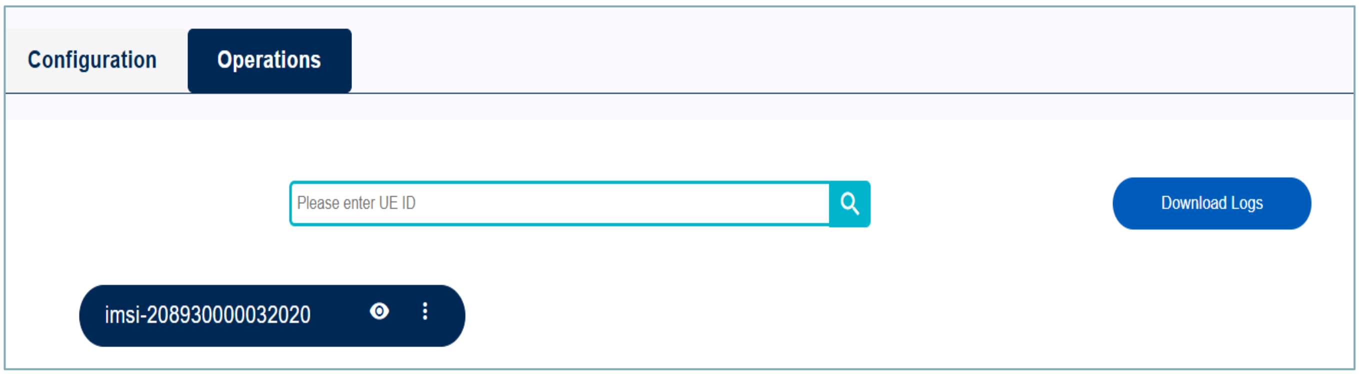

Once UE is registered successfully, "Edit", "Delete" and "Register" buttons gets deactivated, and color is changed to indicate that the registration is successful. Also, for every registered UE, an entry will be created in the "Operations" section.


! For successful registration, UE details should be configured in the 5G Core.

Please refer to section 3.2.3 for configuration of subscriber details in the 5G core.

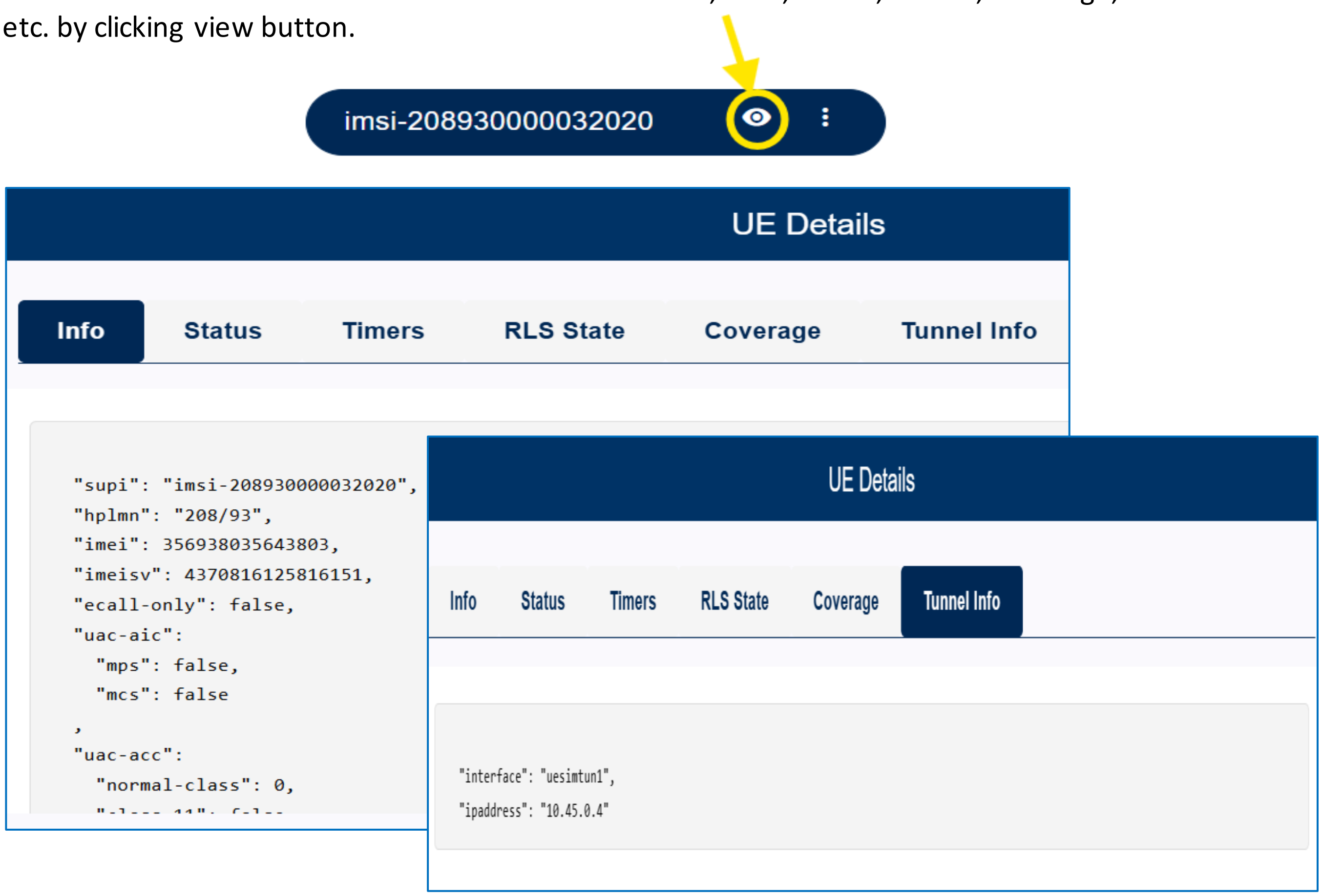
### 4.3.2.2 Operations

Registered UEs will appear in this section. Users can perform various operations like PDU session creation, PDU session release, De-register on the registered UEs.




#### 4.3.2.2.1 View Information

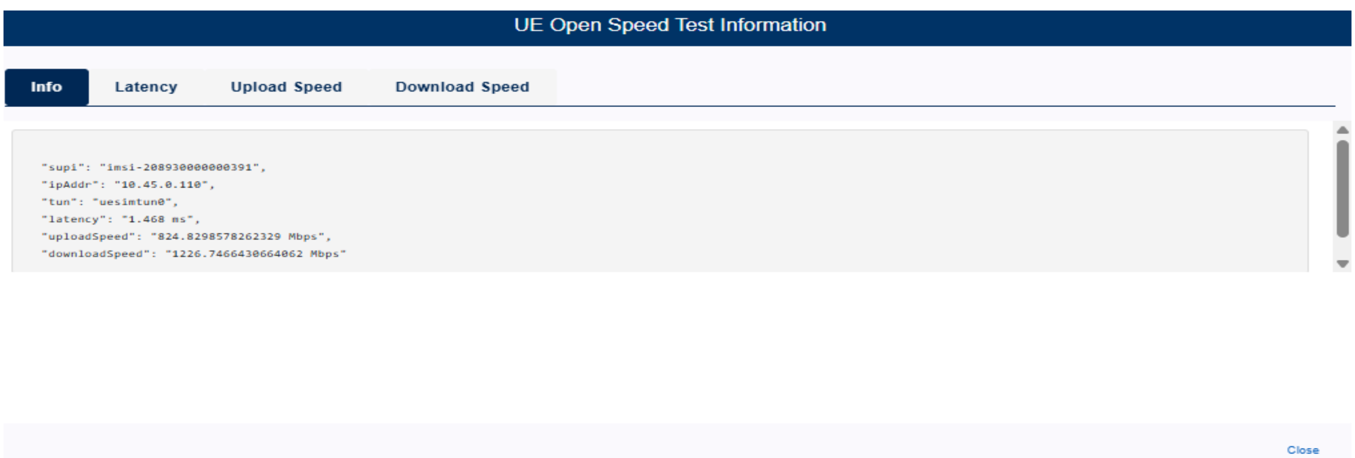
Users can view various information of the UE like PLMN, IMEI, Status, Timers, Coverage, Tunnel details etc. by clicking view button.



### 4.3.2.2.2 More actions

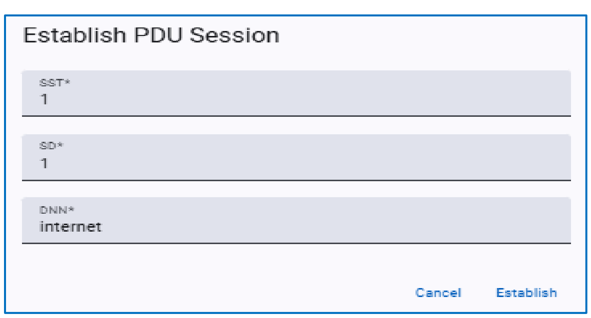
User(s) can perform various operations on the registered UEs using "More actions" button.


**Get Open Speed Test Information:** Returns the Uplink and Downlink speed for that UE.

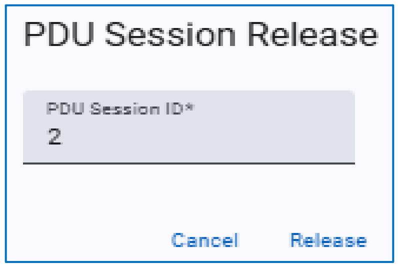



Further detailed information can be seen for Latency, Uplink speed and Downlink Speed.

**Establish PDU Session**: Creates the secondary PDU session

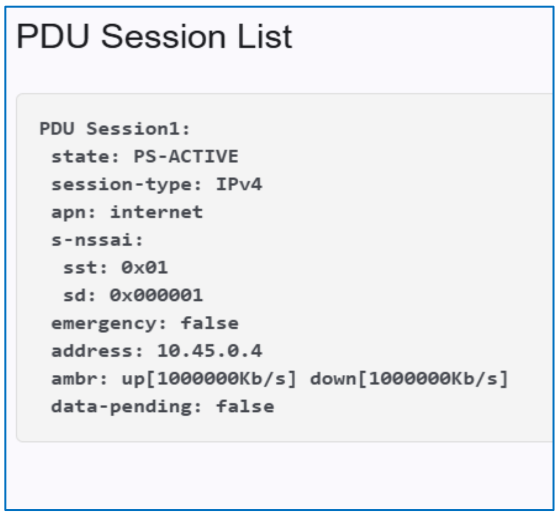


**PDU Session Release:** Release a particular PDU session.




**Release All PDU sessions:** Release all the active PDU sessions for this UE.

**Get PDU Session List:** List all the active PDU sessions of the UE




**Dump all UE Information:** Download all the information of a registered UE in a local text file.

**Deregister:** Perform de-registration of the UE with the 5G Core. UE entry will be removed from the operations section and all buttons will be enabled for this UE in configuration section.

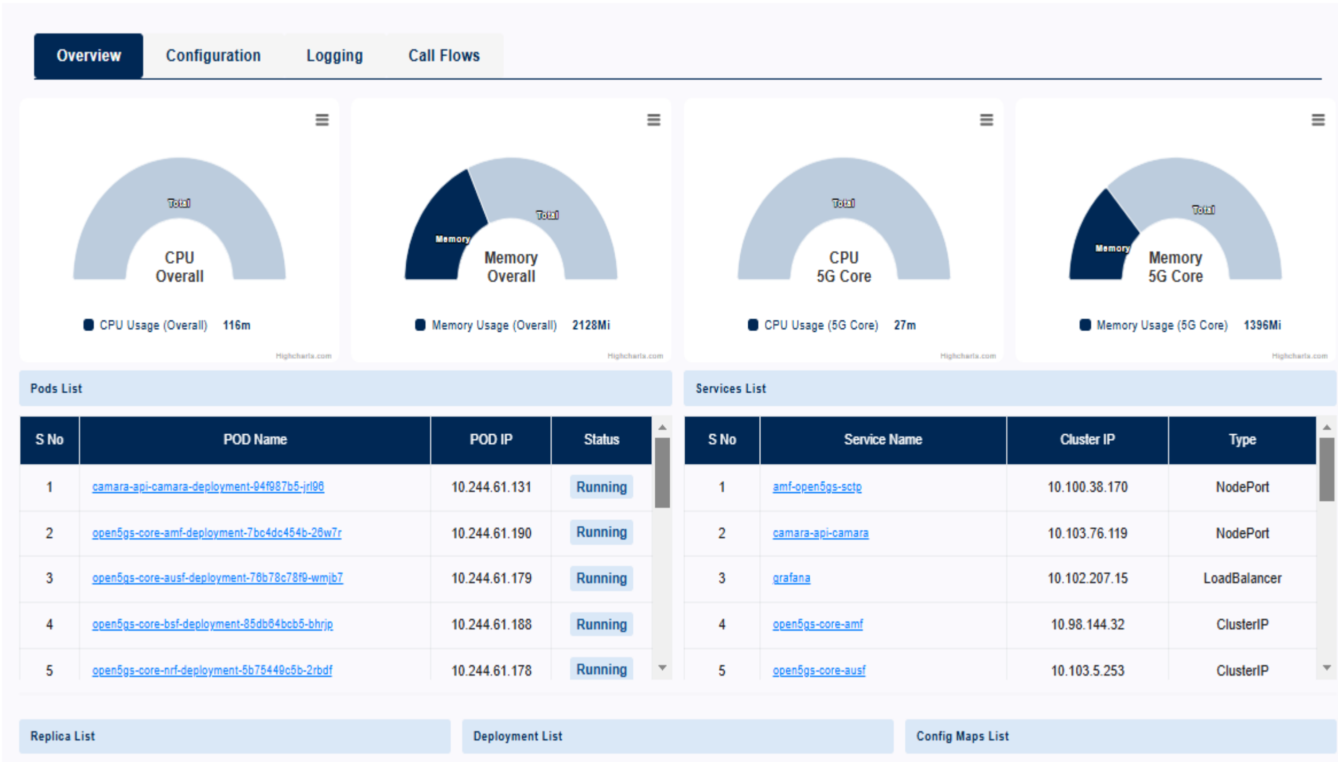
## 4.3.3 5G Core

This section contains the sub-sections below.

- Overview
- Configuration
- Logging
- Call Flows

### 4.3.3.1 Overview

This section provides an overview of the 5G core network including memory usage, CPU usage, pods list, services, deployments and config maps etc.



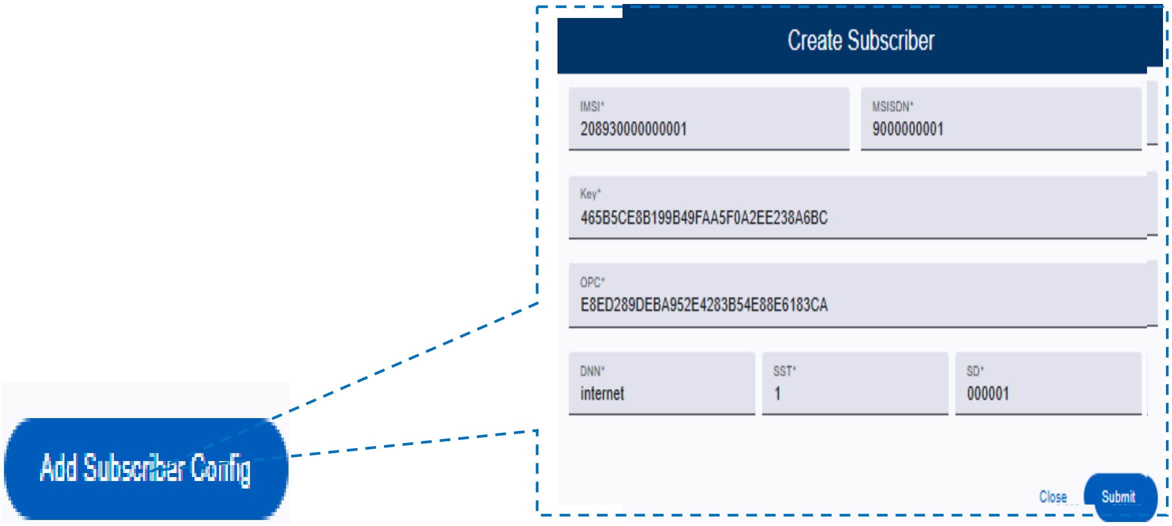

Each 5G network function is orchestrated as Kubernetes pod. Users can view more details of each Kubernetes pod by clicking the pod name.

### 4.3.3.2 Configuration

This section lets users configure the subscribers in the 5G core. Users are provided with three options for the subscriber configuration.

**Option-1: "Add Subscriber configuration using Add"**

- Useful for adding subscribers in bulk.
- Users need to specify the IMSI number, Key Value, OPC, DNN, SST and SD for the subscriber to configured

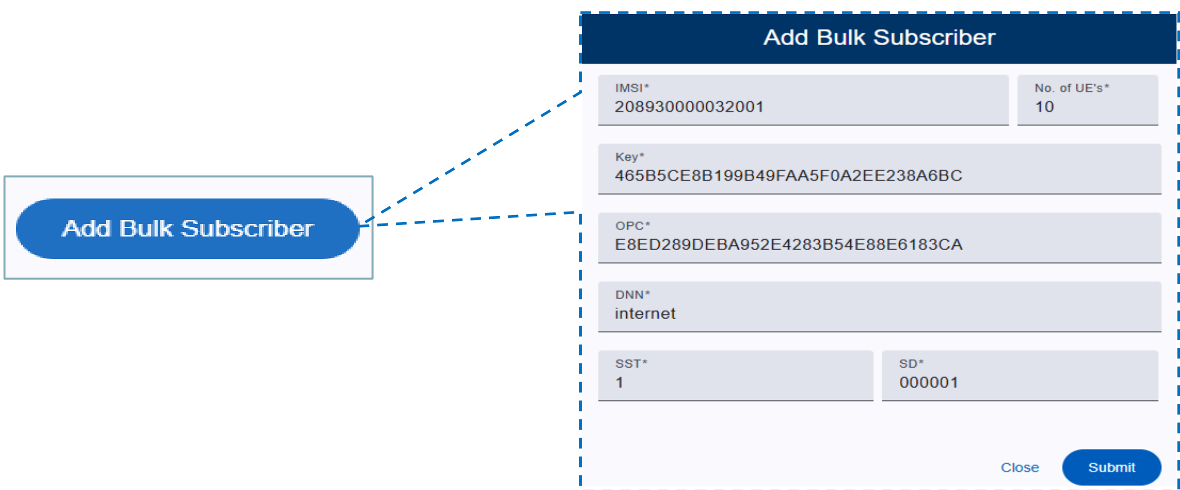



Once the subscriber is configured it will be visible in Testbed UI.

| S No | IMSI | Key | OPC | DNN | SST | SD | MSISDN | Actions |
|---|---|---|---|---|---|---|---|---|
| 1 | 208930000000002 | 465B5CE8B199B49FAA5F0A2EE238A6BC | E8ED289DEBA952E4283B54E88E6183CA | internet | 1 | - | 9876543205 | |
| 2 | 208930000000003 | 465B5CE8B199B49FAA5F0A2EE238A6BC | E8ED289DEBA952E4283B54E88E6183CA | internet | 1 | 000002 | | |
| 3 | 208930000000004 | 465B5CE8B199B49FAA5F0A2EE238A6BC | E8ED289DEBA952E4283B54E88E6183CA | internet | 1 | 000001 | 1234455678 | |

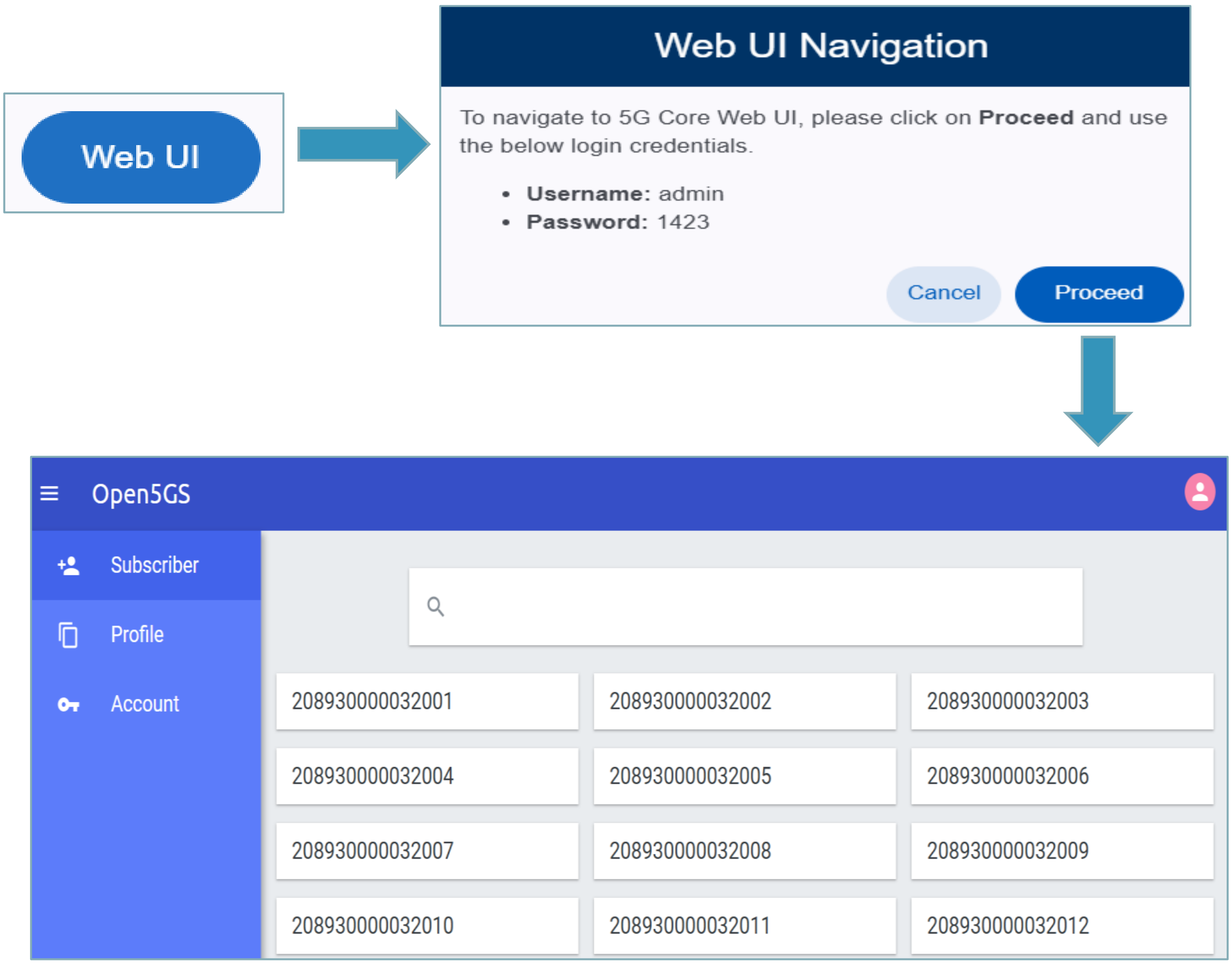
**Option-2: "Add Bulk Subscribers"**

- Useful for adding subscribers in bulk.
- Users need to specify the starting IMSI number and number of subscribers to configured.



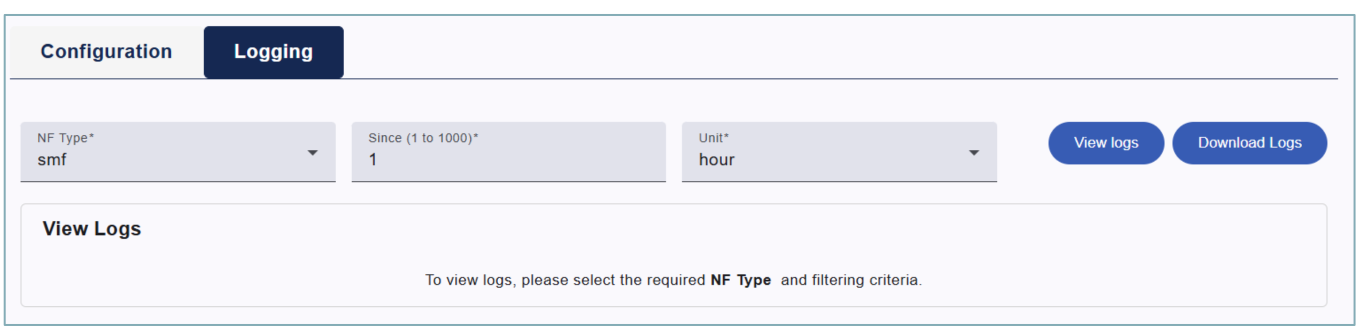
**Option-3: WebUI**

- Users can add, delete and modify subscribers one at a time.



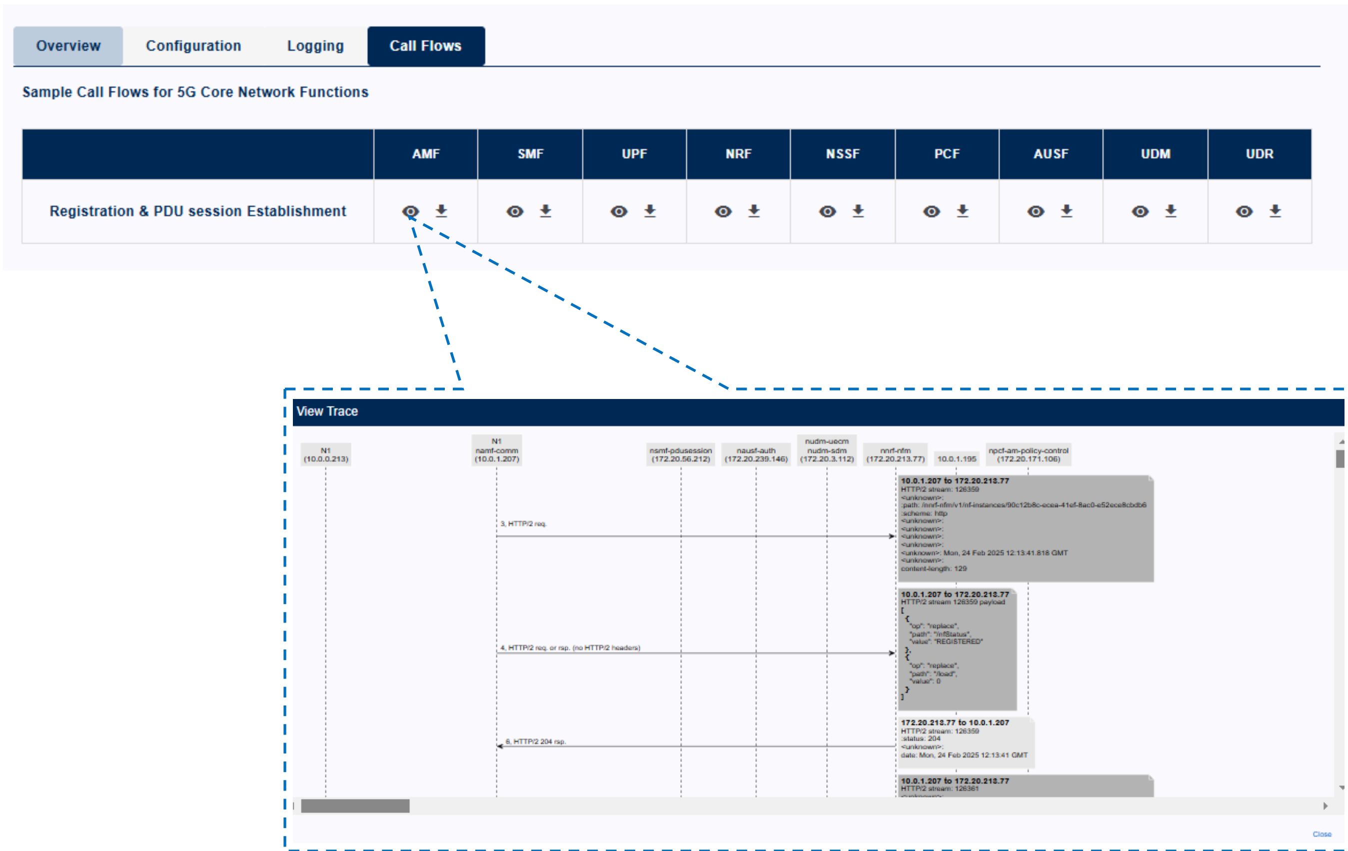

### 4.3.3.3 Logging

This section lets users view or download the logs of a particular 5G core network function.



### 4.3.3.4 Call Flows

This section lets users view or download the sample pcap captures for a particular 5G core network function.




## 4.3.4 Monitoring

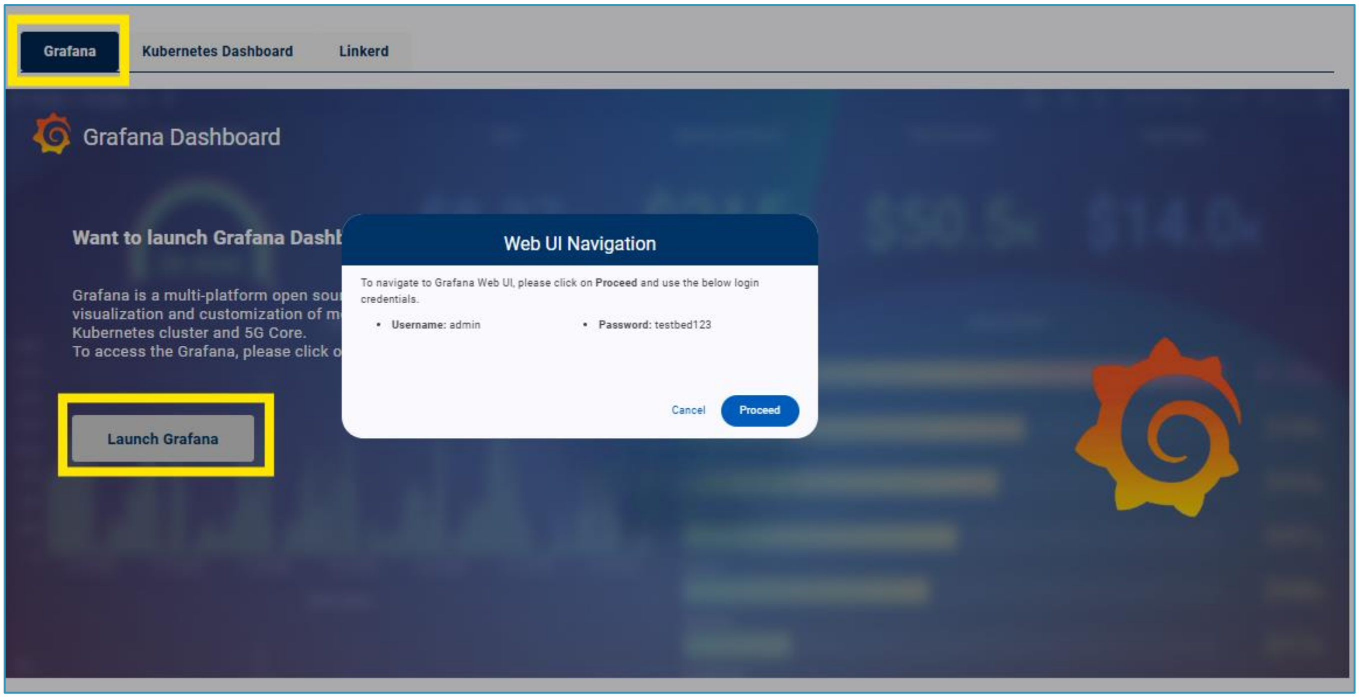
IEEE Testbed provides various tools for monitoring Testbed resources.

- Grafana
- Kubernetes Dashboard
- Linkerd

### 4.3.4.1 Grafana

Users can view the 5G core metrics by using “Launch Grafana” button. If Grafana is not enabled, users can enable it by using “Enable Grafana” button.

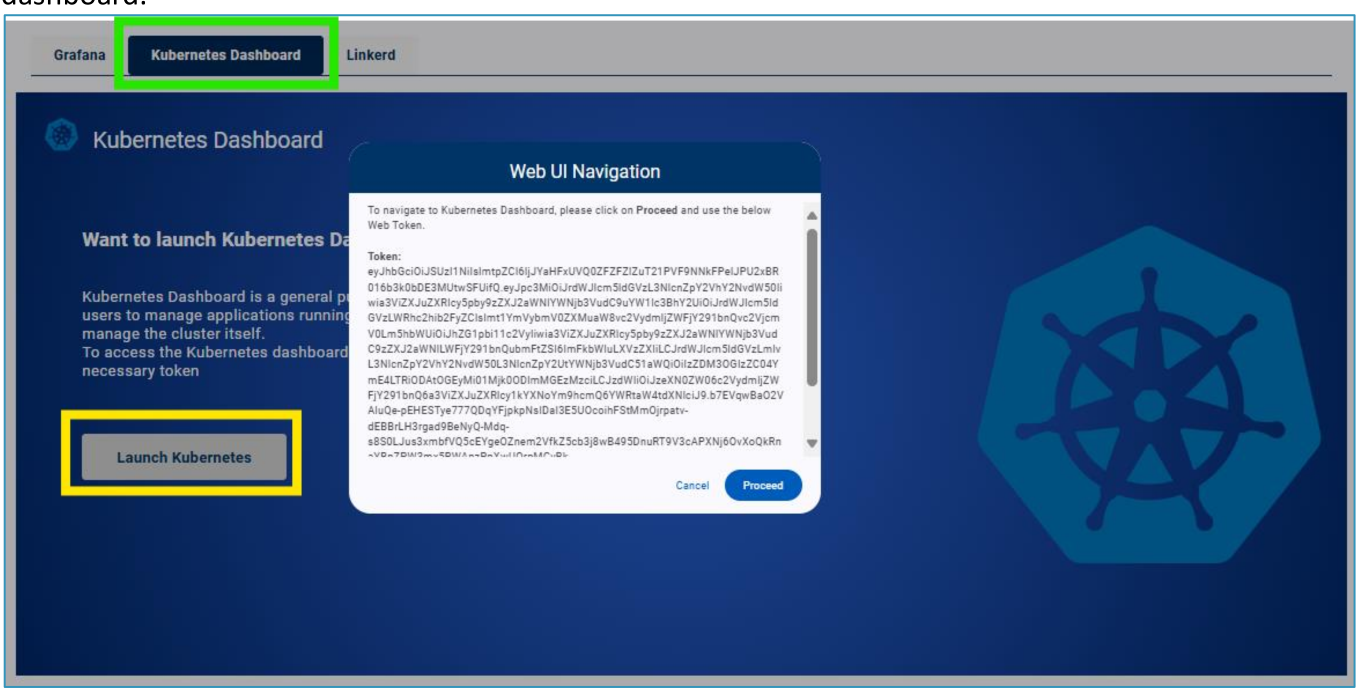
- Username: admin
- Password: testbed123




### 4.3.4.2 Kubernetes Dashboard

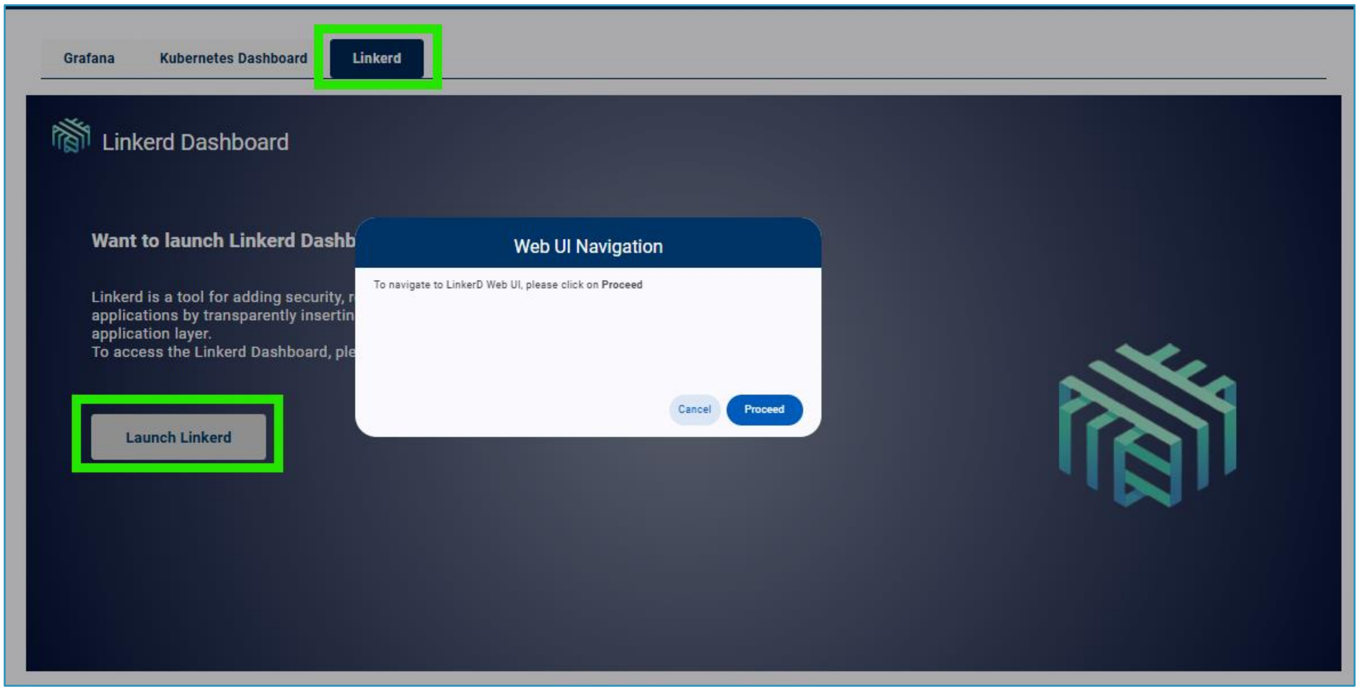
Users can view the detailed Kubernetes dashboard by using “Launch Kubernetes” button. If it's not enabled, user can enable it by using “Enable Kubernetes” button.

Users need to copy the token string and provide it in the Kubernetes window to access the Kubernetes dashboard.



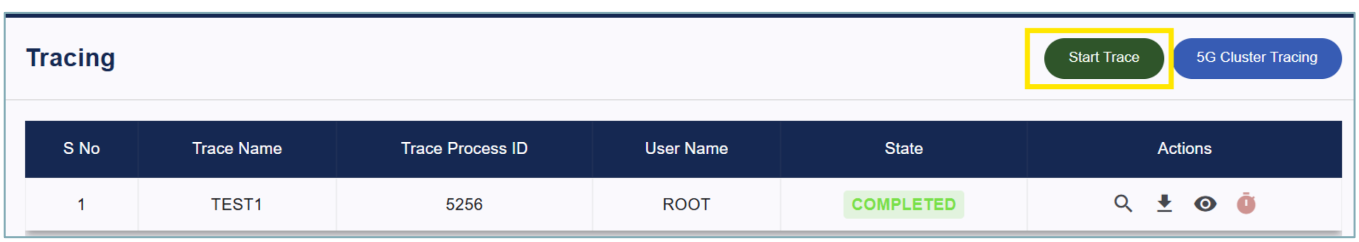
### 4.3.4.3 LinkerD

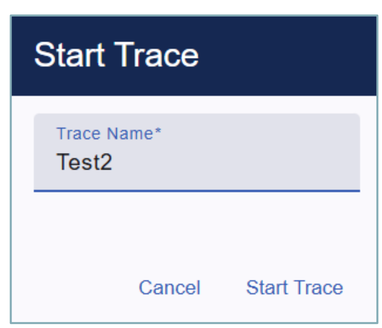
Users can view the detailed LinkerD service mesh by using “Launch LinkerD” button.



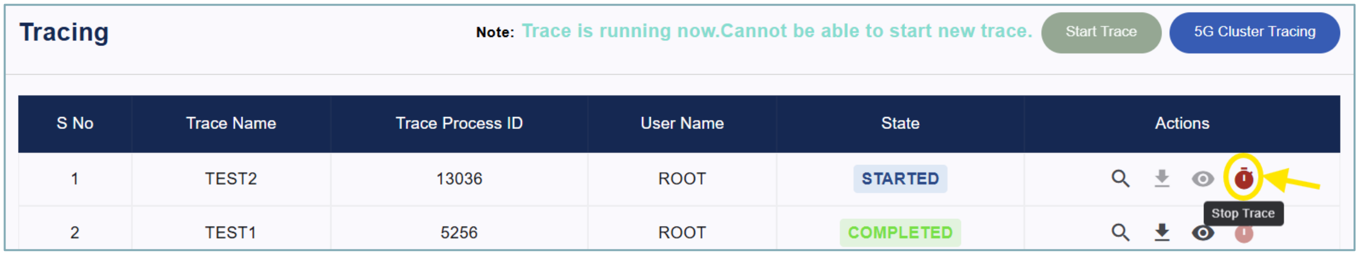

## 4.3.5 Tracing

This section lets the user capture the PCAP traces for their testing.
To start tracing, user need to press “Start trace” button and provide a name.

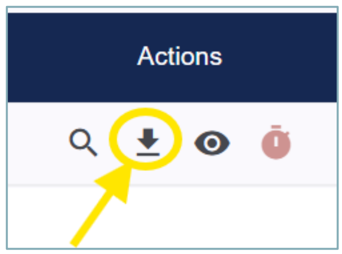


Once testing is complete, user need to stop the tracing by pressing “timer” button.



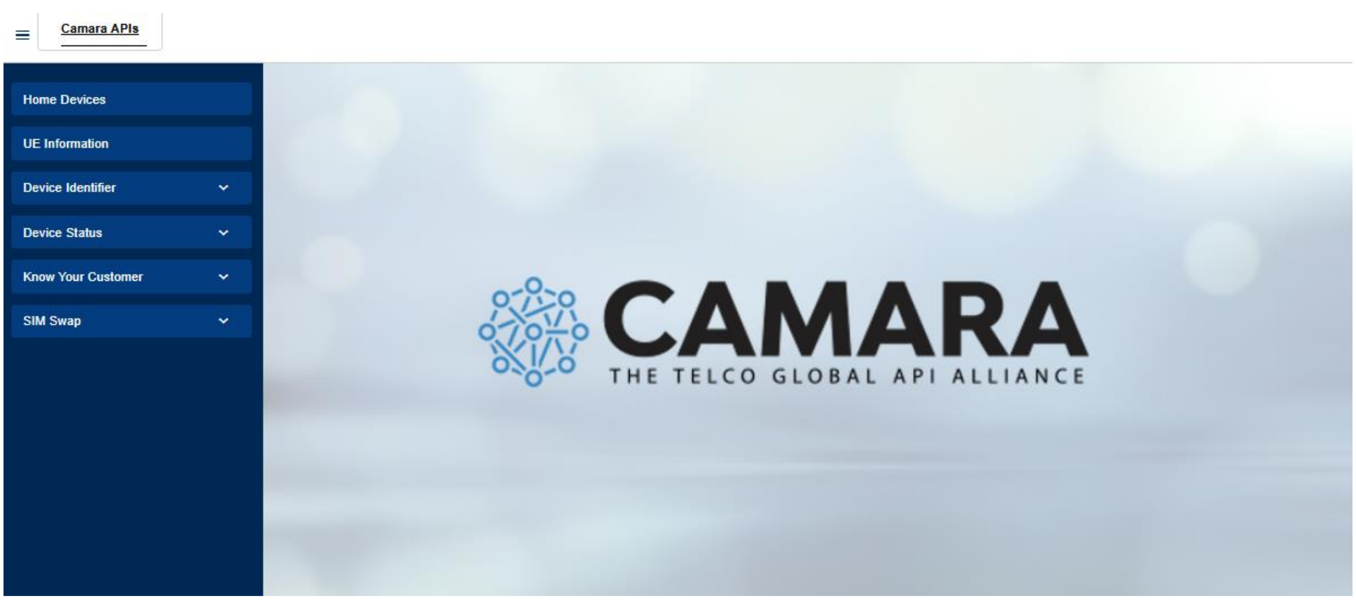

Users can download PCAP logs by clicking download button under Actions column.




# 4.4 Use Cases

## 4.4.1 CAMARA API

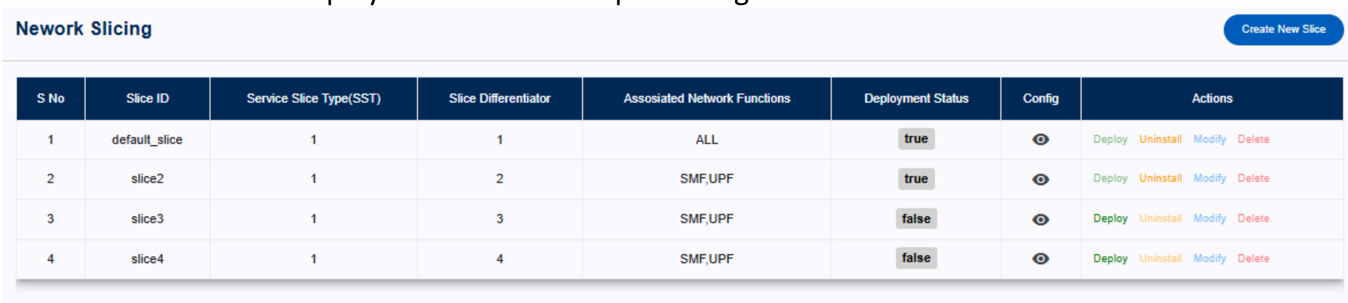
This section lets users view information about the CAMARA API and its use cases and integration with IEEE 5G Testbed.



Please refer to the CAMARA API User Guide for more information

## 4.4.2 Network Slicing

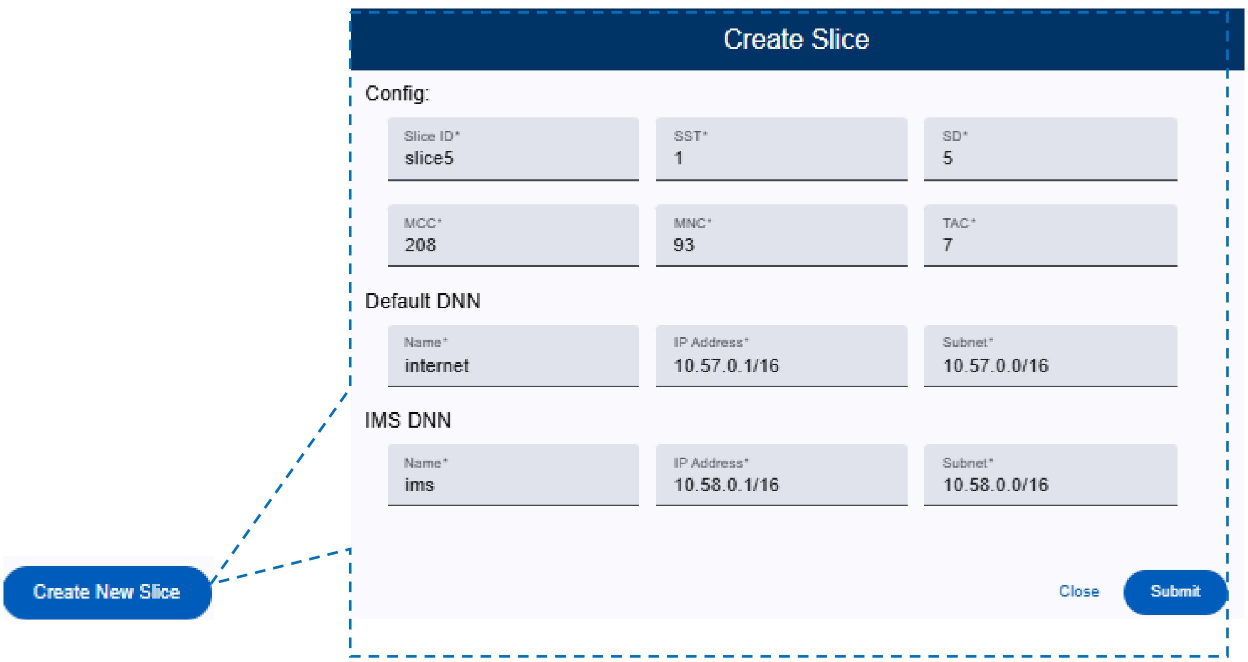
This section lets users deploy and uninstall the pre-configured slices available in the testbed.


Nework Slicing

Create New Slice

| S No | Slice ID | Service Slice Type(SST) | Slice Differentiator | Assosiated Network Functions | Deployment Status | Config | Actions |
|---|---|---|---|---|---|---|---|
| 1 | default_slice | 1 | 1 | ALL | true | | Deploy Uninstall Modify Delete |
| 2 | slice2 | 1 | 2 | SMF,UPF | true | | Deploy Uninstall Modify Delete |
| 3 | slice3 | 1 | 3 | SMF,UPF | false | | Deploy Uninstall Modify Delete |
| 4 | slice4 | 1 | 4 | SMF,UPF | false | | Deploy Uninstall Modify Delete |

User can also create new slices in the Testbed, provides its slicing information is already known to the 5G core.

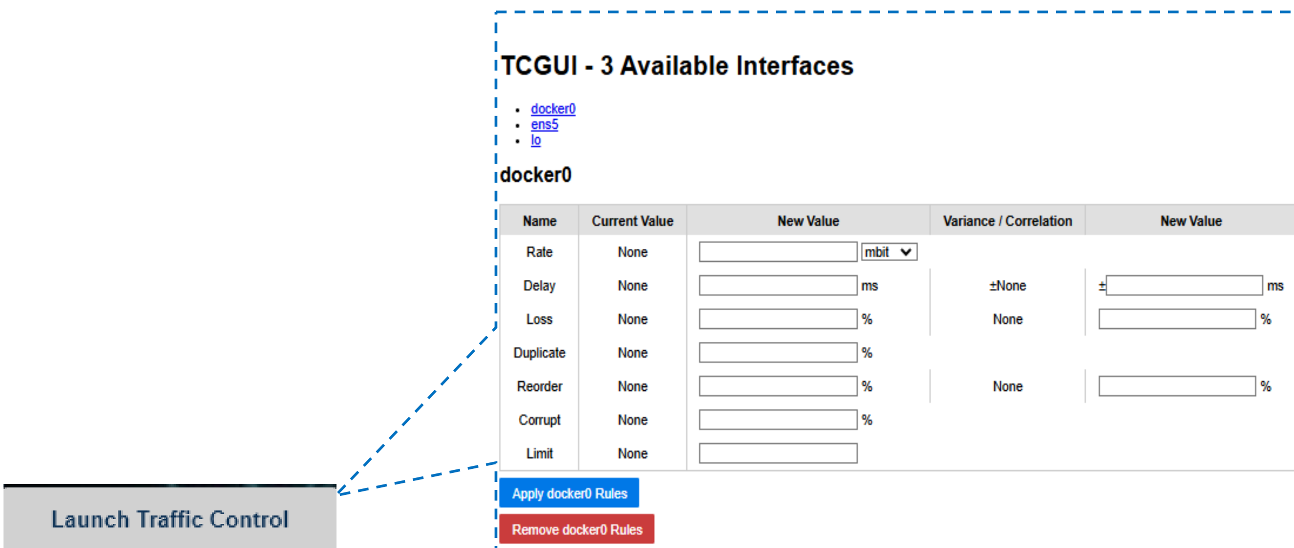


# 4.5 Tools

## 4.5.1 Traffic Control

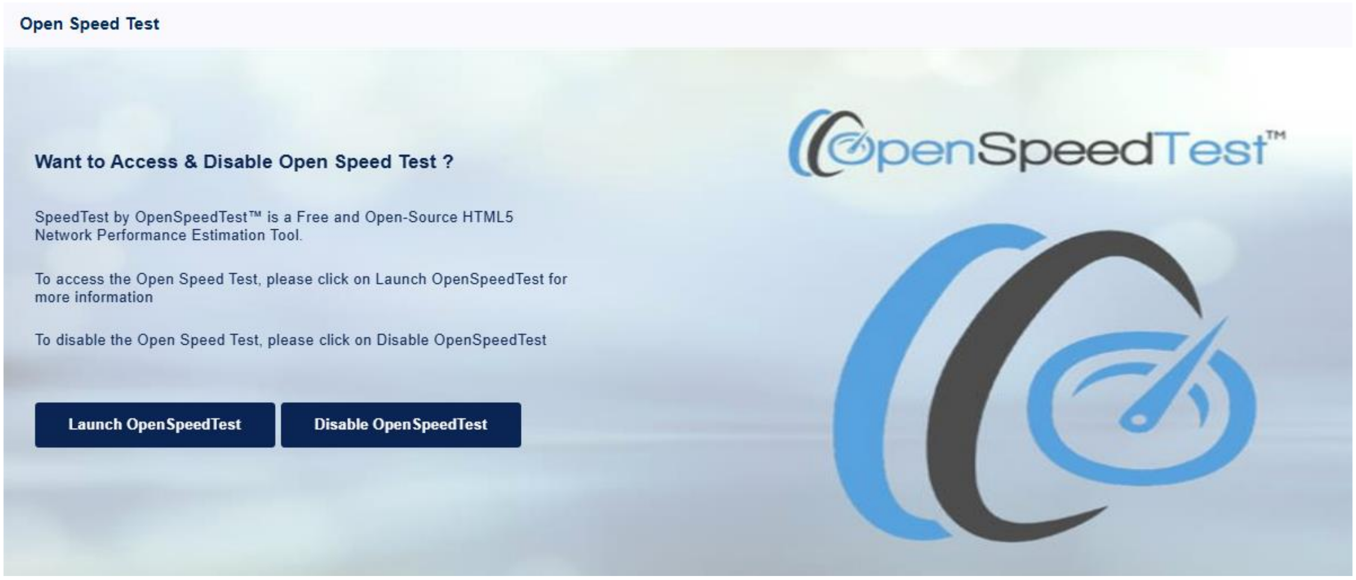
This section lets users to manage and do manipulation of the transmission of packets using the Linux Traffic Control (TC) subsystem helps in policing, classifying, shaping, and scheduling network traffic.


Once user clicks on "Launch Traffic Control", user will be redirected to TCGUI for managing and manipulating the transmission of packets.

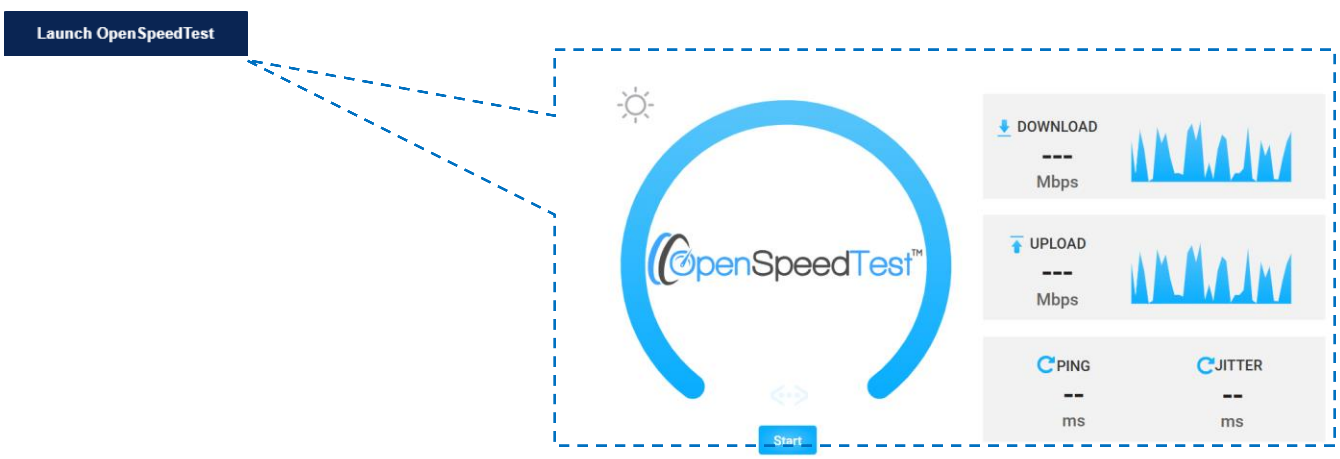


## 4.5.2 Open Speed Test

This section gives user flexibility to enable, disable and access the Open Speed Test – a Network Performance Estimation and Calculation Tool for data rates and latencies.



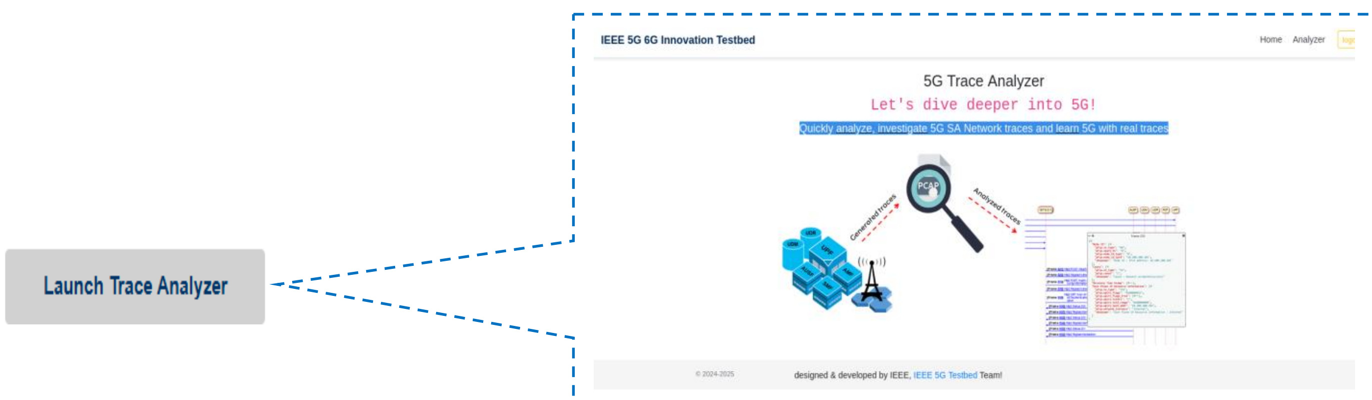

Once user clicks on "Launch Open Speed Test", user will be redirected to Open Speed test GUI performing different kind of speed & latency tests



## 4.5.3 Trace Analyzer

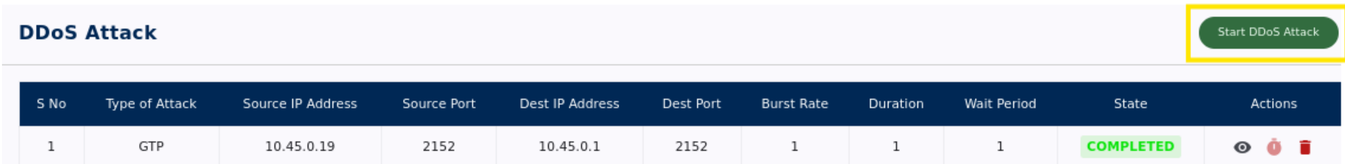
This section gives user flexibility to quickly analyze, investigate 5G SA Network traces and learn 5G with real traces.


Once user clicks on "Launch Trace Analyzer", user will be redirected to Trace Analyzer GUI quickly analyze, investigate 5G SA network traces.



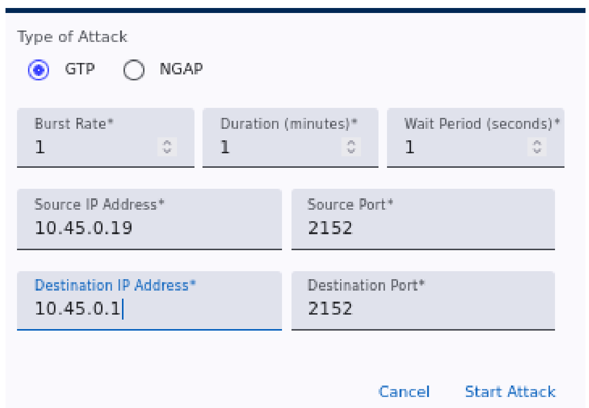
## 4.5.4 DDoS Attack

This section gives users the ability to perform the NGAP and GTP DDoS (Distributed Denial of Service) attacks on Control and Data planes respectively.
To start a DDoS attack, user need to press “Start DDoS Attack” button and provide the required parameters.

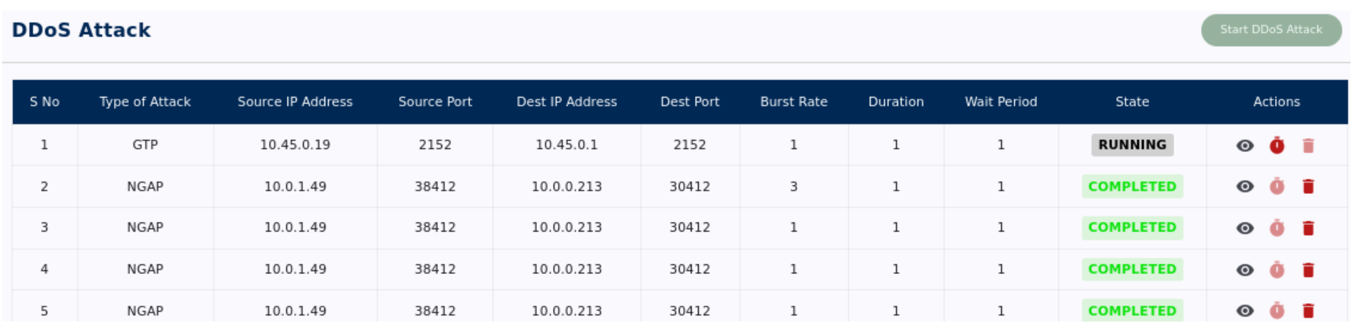


DDoS Attack — Start DDoS Attack

| S No | Type of Attack | Source IP Address | Source Port | Dest IP Address | Dest Port | Burst Rate | Duration | Wait Period | State | Actions |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | GTP | 10.45.0.19 | 2152 | 10.45.0.1 | 2152 | 1 | 1 | 1 | COMPLETED | |



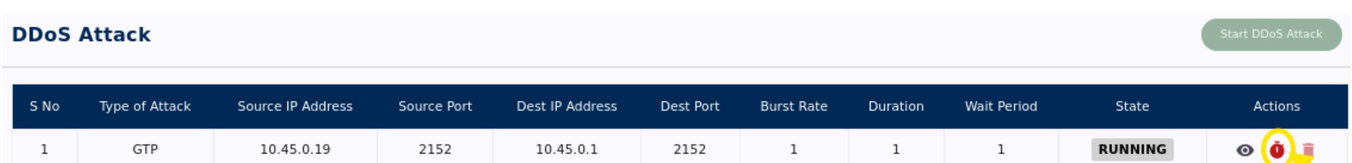

Once the DDoS attack is started, it will be in “RUNNING” state



DDoS Attack — Start DDoS Attack

| S No | Type of Attack | Source IP Address | Source Port | Dest IP Address | Dest Port | Burst Rate | Duration | Wait Period | State | Actions |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | GTP | 10.45.0.19 | 2152 | 10.45.0.1 | 2152 | 1 | 1 | 1 | RUNNING | |
| 2 | NGAP | 10.0.1.49 | 38412 | 10.0.0.213 | 30412 | 3 | 1 | 1 | COMPLETED | |
| 3 | NGAP | 10.0.1.49 | 38412 | 10.0.0.213 | 30412 | 1 | 1 | 1 | COMPLETED | |
| 4 | NGAP | 10.0.1.49 | 38412 | 10.0.0.213 | 30412 | 1 | 1 | 1 | COMPLETED | |
| 5 | NGAP | 10.0.1.49 | 38412 | 10.0.0.213 | 30412 | 1 | 1 | 1 | COMPLETED | |

Once testing is complete, user need to stop the tracing by pressing “timer” button.



DDoS Attack — Start DDoS Attack

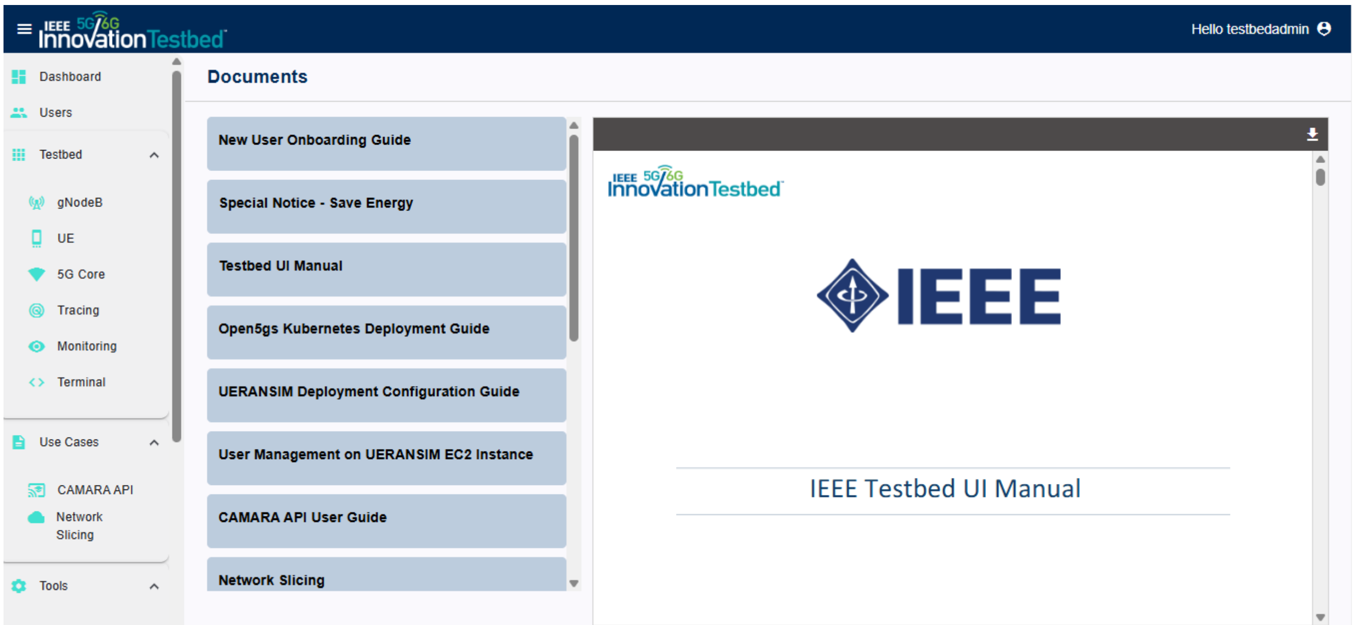
| S No | Type of Attack | Source IP Address | Source Port | Dest IP Address | Dest Port | Burst Rate | Duration | Wait Period | State | Actions |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | GTP | 10.45.0.19 | 2152 | 10.45.0.1 | 2152 | 1 | 1 | 1 | RUNNING | |

# 4.5 Resources

This section provides resources that include the list of documents and videos for different uses cases and features available in the Testbed.

## 4.5.5 Documents

This section documents for different uses cases and features available in the Testbed




## 4.5.6 Videos

This section videos for different uses cases and features available in the Testbed